THE LOCUS WHERE A MORPHISM OF ALGEBRAIC SPACES IS FLAT IS OPEN AND BASE-CHANGE STABLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11707-11726, immediately following the statement at 11686-11705.

Editorial note. One same-chapter lemma and one cross-chapter lemma are recorded in assembly.external_references.

Editorial note. The displayed description of the point set of W is retained verbatim.

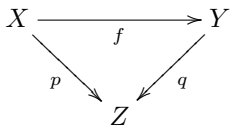
Allowed standard prerequisites (editorial).

- Base change of the flatness locus in fibres. chapter lemma lemma-base-change-flatness-fibres, cited in the proof. Makes the flatness locus A inside |X| open in |X| and shows its formation commutes with arbitrary base change.
- Open subspaces of an algebraic space given by open sets of points. Properties of Spaces, lemma spaces-properties-lemma-open-subspaces, cited at proof lines 11715-11717. Realises the constructed set of points W as an open subspace of Z.

Source reference. Original lemma 05X3: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5917–5917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BZ: focused-seeds/geometry/S9-spaces-where-flat-open/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram*



of algebraic spaces. Assume that

(1) *p is locally of finite presentation,*
(2) *p is flat,*
(3) *p is closed, and*
(4) *q is locally of finite type.*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is flat.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. By Lemma 05X3 the set

$$A = \{x \in |X| : X \text{ flat at } x \text{ over } Y\}.$$

is open in $|X|$ and its formation commutes with arbitrary base change. Let $W \subset Z$ be the open subspace (see Properties of Spaces, Lemma 03BZ) with underlying set of points

$$|W| = |Z| \setminus |p|\,(|X| \setminus A)$$

i.e., $z \in |Z|$ is a point of W if and only if the whole fibre of $|X| \to |Z|$ over z is contained in A. This is open because p is closed. Since the formation of A commutes with arbitrary base change it follows that W works. □